Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/12/2012
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate-, and LET-Response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eablakely@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	UNIVERSITY	**Phone:**	510-486-6595
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	One Cyclotron Road		
PI Address 2:	MS977		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	12/20/2010	**End Date:**	12/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Gap change to delete Cancer 05, per PI (Ed., 12/17/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chang, Polly (SRI International)		
Grant/Contract No.:	NNJ11HA94I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Increased cancer risk remains a primary concern for travel into deep space and may preclude manned missions to Mars due to large uncertainties that currently exist in estimating cancer risk from the spectrum of radiations found in space with available human epidemiological radiation-induced cancer data. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluences of radiations found in space using radiation quality factors and dose-rate modification factors, and assuming linearity of response since the shapes of the dose-responses at low doses below 100 mSv are unknown. The goal of this proposal is to reduce uncertainties in the RBE/LET relationship for particle radiation carcinogenesis. Our approach is to use the existing data on Harderian gland tumorigenesis as reference, and obtain Harderian gland tumor data for heavy ions beams not previously reported to fill the gaps in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our		

	understanding of the dose-response curve at low doses and to test for deviations from linearity. The hypothesis to be tested is that filling in these data gaps will delineate more comprehensively the space radiation risk of space radiation induced tumorigenesis, in particular to provide data that will allow us to test for a role of targeted versus non-targeted effects in cancer risk. We aim to investigate the effects of a broader range of radiation qualities, doses and dose rates not previously reported on the development of Harderian gland and also lung tumors in ~100-120 day old female CB6F1 mice 16 months after exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of the proposed work will fill a significant gap in Harderian tumor response data at low fluences of Silicon or Titanium particle beams to fill in the tumor prevalence in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses relevant to space travel and to test for deviations from linearity. The data should contribute to decreasing the current uncertainty in estimating radiation risk of cancer due to exposures in space travel.
Task Progress:	A Harderian gland tumorigensis study after low doses of 260 MeV/u Silicon ions (LET ~ 70 keV/µm) was completed during this reporting period. The protocol used for this study was modeled as closely as possible to the protocol used by Alpen et al. (1993). All the facilities used for these studies are AAALAC-accredited and the principles promulgated in the NRC Guide for Care and Use of Laboratory Animal Medicine and Animal Welfare Standards are followed at all times. All animal use procedures are approved by each institutional animal use committees. Power calculations were performed to optimize the number of animals in each dose group to obtain results that are statistically meaningful at the relevant low particle radiation doses specified. Harderian gland tissues were examined by a board certified veterinarian histopathologist. Comparative analysis of the historical frequency of Harderian tumors in the same strain of animals for neon and iron ion beams with LETs bracketing the 70 keV/µm of the 260 MeV/u silicon beam clearly show that our recent results show an increase in tumor incidence as a function of low doses and that these results are consistent with the trend established by Alpen et al., 1993. Furthermore, although the animal numbers in the control cohort are still considered low, nevertheless, the spontaneous Harderian gland tumor frequency without the pituitary implants is consistent with the established spontaneous background tumor incidence in the earlier work. Analysis work is now in progress to evaluate whether the tumor incidences at low doses of Silicon ions fit into the Non-Targeted Effect (NTE) model. Lung tissues and other tissues with gross abnormalities at low doses of Silicon ions were also examined histologically. In summary, we see a modest increase in the frequency of lung cancers at low doses with a significant increase at 32 cGy. Most of the lung cancers were bronchoalveolar adenomas. A variety of other tumors were recorded with increases in tumor incidences at doses as low as 4 cGy. We are working on establishing collaborations with collaborators in the Lung NSCOR and others to maximize the research use of the tissues collected from this study.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Blakely EA, Chang PY. "Status Update of Harderian Gland Tumorigenesis." Presented at the Lung Cancer Consortium, NSCOR meeting, Durham, North Carolina, July 7, 2012. Lung Cancer Consortium, NSCOR meeting, Durham, North Carolina, July 7, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Chang PY, Bjornstad KA, Rosen CJ, Bakke JP, Wang A, Du N, Colom RM, Blakely EA. "Low-Dose, High LET Harderian Gland Tumorigenesis: A Progress Report." 23rd NASA Annual Space Radiation Investigators' Workshop, Durham NC, July 8-11, 2012. 23rd NASA Annual Space Radiation Investigators' Workshop, Durham NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Chang PY, Blakely EA. "Harderian Gland Tumorigenesis." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Workshop on Questions in Space Radiation Research. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. , Oct-2012
Articles in Peer-reviewed Journals	Blakely EA. "Lauriston S. Taylor Lecture on radiation protection and measurements: what makes particle radiation so effective?" Health Physics. 2012 Nov;103(5):508-28. http://dx.doi.org/10.1097/HP.0b013e31826a5b85 ; PubMed PMID: 23032880 , Nov-2012
Articles in Peer-reviewed Journals	Chang PY. "Introduction of the Thirty-fifth Lauriston S. Taylor Lecture in Radiation Protection and Measurements." Health Physics. 2012 Nov;103(5):505-7. http://dx.doi.org/10.1097/HP.0b013e318265a23e , Nov-2012